

# SE YOON LEE, PH.D.

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<https://sites.google.com/view/seyoonlee>

## EDUCATION

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**Ph.D. Statistics, Texas A&M University, College Station, U.S.A.**

*May 2021*

Advisor: Prof. Bani K. Mallick

Overall GPA: 3.78/4

**M.A. Applied Statistics, Yonsei University, Seoul, South Korea**

*Feb 2016*

Advisor: Prof. Joseph H.T. Kim

Overall GPA: 3.86/4

**B.S. Mathematics, Yonsei University, Seoul, South Korea**

*Feb 2013*

Graduated with the 2nd highest class rank out of 43 students

Overall GPA: 3.84/4

## RESEARCH INTEREST

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Causal Inference, Semiparametric Efficiency Theory, Bayesian Analysis, Clinical Trials, Biostatistics, ML/AI

## DISSERTATION AND THESIS

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- “Bayesian Hierarchical Modeling: Application towards Complex and High-dimensional Data,” *Texas A&M University*, Doctoral Dissertation ; [\[Link\]](#)
- “Exponentiated generalized Pareto distribution: an alternative to the generalized Pareto distribution,” *Yonsei University*, Master’s Thesis ; [\[Link\]](#)

## WORK EXPERIENCE

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**AbbVie Inc.**

*Feb 2025 – present*

**Manager, Statistics (Permanent Employment)**

North Chicago, IL

- Conducted research on modern causal inference methods for augmenting clinical trials with real-world data (RWD).
- Developed advanced survival analysis methods in causal–RWD settings.
- Supported and led various studies within the RWD group (e.g., observational studies, Health Technology Assessment submissions).
- Actively established myself as a causal inference expert (grounded in semiparametric efficiency theory and ML/AI) with strong domain knowledge in the pharmaceutical industry and academia by participating in conferences and publishing in renowned journals.

**Edwards Lifesciences.**

*May 2023 – Feb 2025*

**Manager, Biostatistics (Permanent Employment)**

Irvine, CA

- Developed an R code to perform unblinded sample size re-estimation (promising zone approach) using CHW test statistics for a hierarchical composite endpoint, employing the Finkelstein and Schoenfeld methodology.
- Specialized in developing simulation R code for power calculation of Bayesian adaptive design for cardiovascular trials at the regulatory submission level, with expertise in end-to-end coding and Statistical Analysis Plan writing.
- Provided seminars on the topic of Bayesian Medical Device Trials to train biostatisticians ; [\[Link\]](#)
- Provided advanced biostatistical supports (e.g., power analysis, primary endpoint analyses, clinical trial designs, etc) to cross-functional teams.

**Johnson & Johnson.***Apr 2022 – May 2023***Senior Biostatistician (Permanent Employment)**

Irvine, CA

- Developed Bayesian adaptive clinical trials designs to evaluate the safety and effectiveness of medical devices (Bayesian basket trial designs, power-prior designs, Bayesian group sequential designs, etc).
- Provided advanced biostatistical supports (e.g., power analysis, primary endpoint analyses, clinical trial designs, etc) to clinicians with main focus on cardiovascular diseases.
- Led working groups of experts (Machine Learning/Statistical Methodology & Consulting) as leader
- Led the development of basket trial designs and authored Statistical Analysis Plan for SECURE, a Post-Market Study [[ClinicalTrials.gov ID: NCT04750798](#)]

**Amgen Inc.***May 2021 – Apr 2022***Scientist - Modeling & Simulations (Permanent Employment)**

Thousand Oaks, CA

- Developed Pharmacokinetics/Pharmacodynamics/Cox hazard regression models for Phase I cancer clinical trials of AMG160 to treat a metastatic castration-resistant prostate cancer (mCRPC).
- Researched machine learning and deep neural network models to fit single dose data for subsequent simulation of multiple dosing scenarios.

**Novartis International AG.***May 2020 – Aug 2020***Biostatistics Summer Intern (Contract Employment)**

East Hanover, NJ

- Developed a Bayesian linear mixed effect model for patients who have wet age-related macular degeneration to predict best-corrected visual acuity over the maintenance phase and suggest a personalized dose regimen. The proposed model has been trained by actual patients' data from HAWK and HARRIER studies (Number of patients is around 1,800 patients). ; [[Abstract](#)]

**EMD Serono Inc. Merck KGaA.***May 2019 – Aug 2019***Pharmacometrics Summer Intern (Contract Employment)**

BillERICA, MA

- Developed a Bayesian adaptive clinical trial design in Phase I cancer clinical trials, which aimed at utilizing grade information from the Common Toxicity Criteria for Adverse Events provided by the National Cancer Institute. ; [[Abstract](#)]  
; [[Poster](#)]

**Texas A&M University.***Aug 2016 – May 2021***Graduate Research/Teaching Assistant**

College Station, TX

- Researched on applications and developments of various statistical models (e.g. non-linear mixed effect model, nonparametric/semiparametric longitudinal model, clustering analysis, classification, hierarchical Poisson model, etc) to various industrial problems arising from biomedical, petroleum, wind energy industries, and COVID-19 outbreak.
- Served as a Teaching Assistant for both undergraduate and graduate courses.

**PUBLICATIONS**

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**Journal Article (\*: corresponding author)**

- [1] **Seyoon Lee**, Joseph H.T. Kim\*. (2018) "Exponentiated generalized Pareto distribution: Properties and applications towards extreme value theory," *Communications in Statistics - Theory and Methods*, 48:8, 2014-2038
- [2] **Se Yoon Lee**, Bowen Lei, and Bani K. Mallick. (2020) "Estimation of COVID-19 spread curves integrating global data and borrowing information," *PLOS ONE*; [[Github](#)]
- [3] Kahkashan Afrin\*, Ashif Iquebal\*, Mostafa Karimi\*, Allyson Larsen\*, **Se Yoon Lee**\*, and Bani K Mallick\*,\*. (2020) "Directionally Dependent Multi-View Clustering Using Copula Model," *PLOS ONE* (\* : equal contribution, authors are alphabetically ordered in the last name.)
- [4] **Se Yoon Lee**\* and Bani K. Mallick. (2021) "Bayesian Hierarchical modeling: application towards production results in the Eagle Ford Shale of South Texas," *Sankhyā: The Indian Journal of Statistics, Series B* ; [[Github](#)]

- [5] **Se Yoon Lee\***. (2021) “Gibbs sampler and coordinate ascent variational inference: a set-theoretical review,” *Communications in Statistics - Theory and Methods*
- [6] **Se Yoon Lee**, Alain Munafo, Pascal Girard, and Kosalarame Goteti\*. (2022) “Optimization of dose selection using multiple surrogates of toxicity as a continuous variable in Phase I cancer trial,” *Contemporary Clinical Trials*; [\[Github\]](#)
- [7] **Se Yoon Lee\***. (2022) “Bayesian Nonlinear Models for Repeated Measurement Data: An Overview, Implementation, and Applications,” *Mathematics*
- [8] **Se Yoon Lee\***. (2022) “The Use of a Log-Normal Prior for the Student t-Distribution,” *Axioms*
- [9] Abhinav Prakash, **Se Yoon Lee**, Xin Liu, Lei Liu, Bani Mallick, Yu Ding\*. (2023) “A Bayesian hierarchical model to understand the effect of terrain on wind turbine power curves,” *IEEE Transactions on Sustainable Energy*
- [10] **Se Yoon Lee\***. (2023) “A flexible dose-response modeling framework based on continuous toxicity outcomes in phase I cancer clinical trials,” *BMC Trials*
- [11] **Se Yoon Lee\***. (2024) “Using Bayesian Statistics in Confirmatory Clinical Trials in the Regulatory Setting: A Tutorial Review,” *BMC Medical Research Methodology*
- [12] **Se Yoon Lee\***. (2024) “Eliciting the discount parameter in a power prior method on the basis of the type I error consideration,” *Statistics in Biopharmaceutical Research*
- [13] **Se Yoon Lee\***, Peng Zhao, Debdeep Pati, Bani K. Mallick. (2024) “Tail-adaptive Bayesian Shrinkage”, *Electronic Journal of Statistics*
- [14] **Se Yoon Lee\***. (2025) “A note on the sample size formula for a win ratio endpoint”, *Statistics in Medicine*
- [15] **Se Yoon Lee\***. (2025) “Power priors and type I error control: constrained borrowing of external control data”, *Journal of Biopharmaceutical Statistics*
- [16] **Se Yoon Lee\***. (2026) “Asymptotic validity of Schoenfeld’s sample size formula for the Cox proportional hazards model via the Wald test approach”, *Statistical Methods in Medical Research*
- [17] **Se Yoon Lee\*** and Jae Kwang Kim. (2026) “MEC: Machine-Learning-Assisted Generalized Entropy Calibration for Semi-Supervised Mean Estimation”, *Forty-third International Conference on Machine Learning (Just Accepted; Among 23,918 reviewed submissions, of which 6,352 were accepted, corresponding to an acceptance rate of 26.6%)*

### Under Review

- [1] **Se Yoon Lee\***. (2026+) “Reliable Causal Inference for Externally Controlled Single-Arm Trials under Transportability Violations,” *Under Review (contact me for the preprint)*

## PROFESSIONAL ACTIVITIES

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### Editorial Board Member

- BMC Medical Research Methodology (Impact Factor: 3.9)

### Invited Peer Reviewers

- Neural Information Processing Systems (h5-index : 371)
- International Conference on Machine Learning (h5-index : 272; Chosen as a Gold Reviewer)
- Artificial Intelligence Review (Impact Factor : 8.139)
- Nature Medicine (Impact Factor : 58.7)
- Statistics in Medicine (Impact Factor: 2.373)
- Biostatistics (Impact Factor: 1.8)
- Pharmaceutical Statistics (Impact Factor: 1.77)
- Statistical Methods in Medical Research (Impact Factor: 1.6)
- Bayesian Analysis (Impact Factor: 3.728)
- Contemporary Clinical Trials Communications (Impact Factor: 1.4)
- European Journal of Clinical Investigation (Impact Factor: 3.481)
- The Journal of Clinical Pharmacology (Impact Factor: 3.126)
- Frontiers in Public Health (Impact Factor: 3.709)

- The R Journal (Impact Factor: 3.984)
- Computational Geosciences (Impact Factor: 2.948)
- Journal of Applied Statistics (Impact Factor: 1.416)

## HONORS AND AWARDS

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|--------------------------------------------------------------------------------------|-----------------------|
| <b>Travel fund for an invited talk at Yonsei University</b>                          | <i>Dec 2019</i>       |
| Given to an invited speaker for the presentation                                     |                       |
| <b>Travel fund for an invited talk at University of Michigan</b>                     | <i>Jun 2019</i>       |
| Given to an invited speaker for an annual meeting about wind energy                  |                       |
| <b>Travel fund for poster presenter for the Houston Geological Society</b>           | <i>Mar 2018</i>       |
| Given to a poster presenter                                                          |                       |
| <b>Anant Kshirsagar fellowship</b>                                                   | <i>Jul 2018</i>       |
| Given to graduate students who demonstrate excellence in making progress in research |                       |
| <b>The 1st place prize in SETCASA poster session in 2018</b>                         | <i>Apr 2018</i>       |
| Given to only one winner for the poster session                                      |                       |
| <b>Graduate student travel award for JSM</b>                                         | <i>Jul 2017, 2018</i> |
| Given to graduate students who make a presentation at the conference                 |                       |

## SOFTWARE

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### Software used in papers

- Bayesian Hierarchical Richards Model ; Written in R ; [\[Download\]](#) ; [\[Github\]](#)
- Spatial Weibull Model ; Written in R ; [\[Github\]](#)
- bayesestdft ; Written in R ; [\[Github\]](#)

## SKILLS

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| <b>Language</b>          | Fluent in English and Korean; Elementary in Chinese and Japanese |
| <b>Computer Language</b> | Proficient in R, Python, SQL, Microsoft Access, SAS, and NONMEM  |
| <b>Certificate</b>       | Cognigen NONMEM Workshop                                         |

## REFERENCES

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|------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Bani K. Mallick</b> | Distinguished Professor, Texas A&M University, <a href="mailto:bmallick@stat.tamu.edu">bmallick@stat.tamu.edu</a> |
| <b>Debdeep Pati</b>    | Professor, University of Wisconsin - Madison, <a href="mailto:dpatiz@wisc.edu">dpatiz@wisc.edu</a>                |
| <b>Samiran Sinha</b>   | Professor, Texas A&M University, <a href="mailto:sinha@stat.tamu.edu">sinha@stat.tamu.edu</a>                     |
| <b>Alan Dabney</b>     | Professor, Texas A&M University, <a href="mailto:adabney@stat.tamu.edu">adabney@stat.tamu.edu</a>                 |